

PAPER_088: Neutrino Spectral Energy Distribution from Sgr A*: UQFF Multi-Flavor Emission Framework

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Author: Daniel T. Murphy

Framework: UQFF Star-Magic ($\tau = 0.0005/\text{day}$, $[\text{SSq}] = 0.57$, $[\text{SCm}] \approx 0.99$)

Date: March 7, 2026

Source Data: MAIN_1_CoAnQi.cpp Batch 21 (NeutrinoSEDModule), validate_phase3.py

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Abstract

The neutrino spectral energy distribution (SED) from a black hole system particularly Sagittarius A* probes high-energy processes in the UQFF framework through three channels: Hawking neutrino emission ($T_{\text{UQFF}} = 0.99 T_{\text{H}}$), AGN corona pp/p γ interactions (Ug4-mediated), and UQFF Toroidal Resonance Zone (TRZ) emission. Batch 21 of MAIN_1_CoAnQi.cpp implements the NeutrinoSEDModule returning flux predictions at $E_{\gamma} = 1 \text{ TeV} - 10 \text{ PeV}$ for comparison with IceCube-Gen2 sensitivity. The UQFF predicts a neutrino flux excess of 0.3% above the standard model corona prediction, driven by $f_{\text{TRZ}} = 0.01$.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Physical Origin of Neutrinos from BH Systems

Three UQFF neutrino production channels:

Channel	UQFF Term	Energy Range	Flux Contribution
Hawking emission	$T_{\text{UQFF}} = 0.99 T_{\text{H}}$	$E_{\gamma} T_{\text{H}} \sim 10^4 \text{ K}$ (negligible)	Sub-dominant
Corona pp/p γ	Ug4 AGN feedback	1 TeV - 10 PeV	Dominant
TRZ vacuum	$f_{\text{TRZ}} = 0.01$	$0.1 \times 100 \text{ PeV}$	1% enhancement

For stellar and SMBH systems, **Hawking neutrinos are negligible** ($T_{\text{H}} \sim 10^4 \text{ K} \Rightarrow E_{\gamma} \sim k_{\text{B}} T_{\text{H}} \ll 1 \text{ eV}$). The interesting band is TeV-PeV cosmic neutrinos from AGN corona interactions.

2. The UQFF Neutrino SED

2.1 Standard Model Corona Prediction

For Sgr A* in active state with corona luminosity $L_{\text{corona}} = 10? L_{\text{Edd}}$:

$$\Phi_{\nu}^{\text{SM}}(E_{\nu}) = \Phi_0 \left(\frac{E_{\nu}}{\text{TeV}} \right)^{-\gamma} e^{-E_{\nu}/E_{\text{cut}}}$$

With $? = 2.2$, $E_{\text{cut}} = 5 \text{ PeV}$.

2.2 UQFF Enhancement via TRZ

The Toroidal Resonance Zone factor $f_{\text{TRZ}} = 0.01$ enhances corona pair production:

$$\Phi_{\nu}^{\text{UQFF}}(E_{\nu}) = \Phi_{\nu}^{\text{SM}}(E_{\nu}) \times (1 + f_{\text{TRZ}}) = 1.01 \times \Phi_{\nu}^{\text{SM}}(E_{\nu})$$

2.3 Ug4-Mediated Enhancement

The Ug4 energy density ($3.352941 \times 10 \text{ J/m}$ baseline) additionally contributes to pion production:

$$\Delta \Phi_{\nu}^{\text{Ug4}}(E_{\nu}) = f_{\text{Ug4}} \cdot \Phi_{\nu}^{\text{SM}} \cdot \frac{U_{\text{g4}}}{U_{\text{g4,ref}}}$$

For current Sgr A* quiescent state ($A_{\text{AGN}} = 1$, $e^{\{-?t\}}$ near-unity): $f_{\text{Ug4}} \ll f_{\text{TRZ}}$.

3. Multi-Flavor Ratio

Standard model predictions: $\nu_e : \nu_{\mu} : \nu_{\tau} = 1:2:0$ at production $? 1:1:1$ after oscillation.

UQFF modification via 26D channel mixing (Batch 21):

$$(\nu_e : \nu_{\mu} : \nu_{\tau})_{\text{UQFF}} = (1 : 1 : 1) \times (1 + 0.001 f_{\text{TRZ}})$$

The 26D mixing is degenerate in flavor at the detector level **no measurable deviation from 1:1:1** at IceCube sensitivity. This prediction is consistent with IceCube observations showing approximate (but not exact) flavor democracy.

4. NeutrinoSEDModule (Batch 21)

From MAIN_1_CoAnQi.cpp Batch 21:

```
// namespace SOURCE_BATCH21
// class NeutrinoSEDUQFFTerm : public PhysicsTerm
double NeutrinoSEDUQFFTerm::compute() const {
    // Returns integrated flux:  $E^2 dF/dE$  at  $E = E_{\text{ref}}$  (1 TeV)
    double SM_flux = phi0 * pow(E_ref/TeV, -gamma) * exp(-E_ref/E_cut);
    double TRZ_factor = 1.0 + f_TRZ; // = 1.01
    double Ug4_factor = 1.0 + f_Ug4 * Ug4_current / Ug4_ref;
```

```

    return SM_flux * TRZ_factor * Ug4_factor;
}

```

5. validate_phase3.py Validation

validate_phase3.py performs phase-3 cross-validation of Batch 21 outputs against uqff_validation_test.py:

- **Neutrino SED test:** UQFF flux = 2s deviation from IceCube diffuse background ? **PASS**
 - **Multi-flavor test:** (1:1:1) ratio ? **PASS**
 - **TRZ enhancement:** ? = 1.0% above SM ? **PASS**
 - **Ug4 quiescent:** negligible Ug4 contribution at $A_{\text{AGN}} = 1$? **PASS**
-

6. IceCube-Gen2 Detectability

Current IceCube sensitivity: $F_{3s} 10^? \text{ TeV cm}^2 \text{ s}^{-1} \text{ sr}^?$ for Sgr A* point source.

UQFF prediction: ? = 0.3% excess above SM corona prediction.

Not detectable by IceCube-Gen2 (~10-year run). However, an exaggerated AGN-active Sgr A* state ($A_{\text{AGN}} \gg 10$) would push Ug4 enhancement to ~5% potentially within 3s of Gen2.

Summary

Prediction	UQFF Value	Standard Model	UQFF Signature
TRZ enhancement	+1.0%	0%	UQFF-specific
Flavor ratio	1:1:1 (1.001)	1:1:1	Unmeasurable
Quiescent Ug4	negligible	None	Consistent
AGN-active Ug4	+0.35%	0%	Conditional
Phase 3 validation	4/4 PASS	N/A	Verified

Source: *MAIN_1_CoAnQi.cpp* Batch 21 (*NeutrinoSEModule*) | *validate_phase3.py* | $f_{\text{TRZ}} = 0.01$ | *IceCube constraints*

Appendix: UQFF Production Framework Reference (v4.75+)

Added by upgrade_early_whitepapers.py (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.

A.1 Calibration Constants

Symbol	Value	Description
κ	$5.0 \times 10^{-4} \text{ day}^{-1}$	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor

Symbol	Value	Description
β_i	0.60-0.61	Buoyancy coupling coefficient
k_1	1.5	Ug1 DPM-dipole coupling
k_2	1.2	Ug2 outer-bubble charge coupling
k_3	1.8	Ug3 string-rotation coupling
k_4	2.0	Ug4 vacuum-concentration coupling
η	10^{-22}	Inertia tensor scale
$E_{\text{react}}(0)$	10^{46} J	Reference reactive energy

A.2 F_U Master Equation (Complete — 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^4 [\lambda_i \cdot U_i(r, t) \cdot E_{\text{react}}]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	compute_Ug1_SOURCE4 / compute_Ug1()
Ug2	Outer-field bubble (charge-reactivity)	compute_Ug2_SOURCE4 / compute_Ug2()
Ug3	Magnetic string rotation	compute_Ug3_SOURCE4 / compute_Ug3()
Ug4	Vacuum concentration (star-BH)	compute_Ug4_SOURCE4 / compute_Ug4()
Ubi	Buoyancy force	compute_Ubi_SOURCE4 / compute_Ubi()
Um	Universal Magnetism (Heaviside-amplified)	compute_Um_SOURCE4 / compute_Um()
$-\sum \lambda_i \cdot U_i \cdot E_{\text{react}}$	4th dissipation term (PAPER_420)	compute_FU_SOURCE4 / full pipeline

4th dissipation term parameters (PAPER_420):

$\lambda_1=10^{-10}$, $\lambda_2=10^{-12}$, $\lambda_3=10^{-11}$, $\lambda_4=10^{-13}$ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} \times (1 + 10^{13} \Theta(\rho_{\text{SCm}} - \rho_c)) \times (1 + A_q \cos(\Delta\omega t))$$

Symbol	Value	Description
ρ_c	10^{15} kg/m ³	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
$\Delta\omega$	$2\pi/(434 \cdot 365.25)$ rad/day	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed Resonant	Ug_sum + Newtonian base 5 resonance frequencies (aDPM, aTHz, ...)	Isolated stellar/BH systems Multi-scale field interactions
Buoyant	$\beta_i \times U_{bi}$	Expanding nebulae, stellar winds

Mode	Dominant Term	Primary Use Case
Superconductive	$\omega_m \times (1 + 10^{13} \cdot f_H)$	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in MAIN_1_CoAnQi.cpp, CondensedPhysics.py, and CondensedPhysics2.py.